MIL-D-43970
27 September 1977
SUPERSEDING
(See 6.6)

MILITARY SPECIFICATION

DRY CLEANING, RECOVERY TUMBLER, FOR SYNTHETIC SOLVENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers synthetic solvent (perchlorethylene) dry cleaning, recovery tumblers.

1.2 Classification. The dry cleaning tumblers shall be of the following minimum sizes as specified (see 6.2):

35-pound (15.9 kg) capacity, solvent recovery tumbler.
50-pound (22.6 kg) capacity, solvent recovery tumbler.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

CC-M-1807 - Motors, Alternating Current, Fractional and Integral Horsepower (500 Horsepower and Smaller)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3510

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QQ-N-281 - Nickel-Copper Alloy Bar, Plate, Rod, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-288 - Nickel-Copper Alloy and Nickel-Copper-Silicon Alloy Castings.

MILITARY

MIL-T-152 - Treatment, Moisture and Fungus Resistant, of Communications, Electronics, and Associated Electrical Equipment.
MIL-L-3153 - Laundry and Dry Cleaning Machinery and Equipment (For Fixed Installations), Preparation for Delivery Of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-461 - Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-1474 - Noise Limits for Army Material.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

FACTORY MUTUAL RESEARCH CORPORATION

Approval Standard - Classes II, III and IV Commercial Dry Cleaning

(Application for copies should be addressed to Factory Mutual Testing Laboratories, 1151 Providence Highway, Norwood, MA 02062.)



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NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

ICS - Industrial Controls and Systems

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 155 East 44th Street, New York, NY 10017.)

UNDERWRITERS LABORATORIES INC. (UL)

UL 664 - Commercial Dry-Cleaning Machines (Class IV)

(Application for copies should be addressed to the Underwriters Laboratories Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingsten Road, Northbrook, IL 60062; 1285 Walt Whitman Road, Melville, LI, NY 11746; 1655 Scott Boulevard, Santa Clara, CA 95050; or 2602 Tampa East Blvd., Tampa, FL 33619.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

Standard No. 32 - Drycleaning Plants (Class IV)
Standard No. 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Stainless Steel and Heat-Resisting Chromium-Nickel Steel Plate, Sheet and Strip.
A 176 - Stainless and Heat-Resisting Chromium Steel Plate, Sheet, and Strip.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

## 3. REQUIREMENTS

3.1 Standard product. The tumblers furnished under this specification shall be the manufacturer's current standard product except for changes necessary to comply with the requirements of this specification.



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3.2 Codes and standards.

3.2.1 NFPA. The tumblers shall conform to NFPA Standard No. 32. The electrical wiring and conduit shall conform to NFPA Standard No. 70.

3.2.2 UL or FM. The tumblers shall comply with Underwriter's Laboratory Standard UL 664 or Factory Mutual approval standard (see 2.2).

3.2.3 NEMA. Motor controllers, switches, relays, and time delays shall conform to performance requirements of NEMA Publication ICS.

3.2.4 Certification. Prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative, that the tumblers meet the requirements specified in 3.2.1, 3.2.2 and 3.2.3, as applicable.

3.2.4.1 NFPA. Acceptable evidence of meeting the requirements of 3.2.1 shall be the manufacturer's certified statement that the tumblers conform to NFPA Standard No. 32 and are wired to conform to NFPA Standard No. 70.

3.2.4.2 UL or FM. Acceptable evidence of meeting the requirements of 3.2.2 shall be the UL label or FM approval mark in the finished machine or certification from an independent testing laboratory acceptable to the contracting officer or his authorized representative that the machine has been tested and conforms to UL 664.

3.2.4.3 NEMA. Acceptable evidence of meeting the requirements of 3.2.3 shall be the manufacturer's certified statement that the motor controllers, switches, relays and time delays, conform to NEMA Publication ICS. A tag or label attached to these items stating these components conform to this standard is acceptable evidence.

3.3 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer for the specified equipment, provided the completed item complies with all provisions of this specification (see 6.4).

3.3.1 Corrosion-resistant metal. Corrosion-resistant metal used in the fabrication of the tumblers shall be corrosion-resisting steel conforming to the requirements of ASTM A 167 or A 176, as applicable, or nickel-copper alloy conforming to QQ-N-281 or QQ-N-288. Aluminum material doors and cleanout covers may be used as an alternative for the tumblers in accordance with commerical practice.

3.3.2 Materials for fasteners. Rivets, bolts, screws, nuts, and washers shall be carbon or alloy steel. Where brass or corrosion-resistant metal is to be fastened the fasteners shall be brass or corrosion-resistant metal respectively. Where dissimilar metals are to be fastened the fasteners shall be corrosion-resistant metal.



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3.3.3 Pipe, tube, fittings and valves. Pipe, tube, valves and fittings used for solvent, water, air and steam lines shall comply with the manufacturer's commercial practice.

3.3.4 Conduit, fittings, and electric wire. Electrical conduit shall be rigid electrical, metallic tubing or flexible conduit in accordance with standard commercial practice. All electrical wire shall be stranded copper heat-resistant grade, thermoplastic insulated. Solid nonstranded wire shall not be used. All conduit fittings and electric wire shall be UL listed or labeled.

3.3.5 Motors. Motors shall be drip proof or splash proof or totally inclosed for operation in a 40°C ambient temperature. Motors shall conform to the applicable requirements of CC-M-1807 and to the following:

(a) All motors shall have windings impregnated to resist moisture.
(b) Motors shall be equipped with ball bearings, except that motors of 1/2 horsepower (373 W) or less when used for horizontal applications may have sleeve bearings.
(c) Motor bearings shall be of the permanently lubricated type or shall have adequate and accessible means for lubrication.
(d) Each motor of 1/8 horsepower (93 W) rating or larger shall be provided with manually reset thermal-overload protection incorporated as a part of the motor or installed in the controller.

3.4 Design. Design of tumblers shall conform to the requirements specified herein.

3.4.1 Construction. The tumbler shall be for both manual and automatic operation and shall be furnished complete with motors, driving mechanism, starters, and controllers. Master switches, timers, reversing mechanism, and other electrical components of the manufacturer's standard product shall be furnished as required. The machines, when tested as specified in 4.3.1, shall comply with the following:

(a) The assembly of components shall have complete and proper interrelationship.
(b) Moving parts and bearings shall not overheat.
(c) Starting and stopping devices shall operate smoothly and positively.
(d) There shall be no structural failures.



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3.4.1.1 Electrical system.

3.4.1.1.1 Electrical characteristics. The electrical components of the equipment shall be designed for operation on the voltage, frequency, and number of phases as specified (see 6.2).

3.4.1.1.2 Wiring and instructions. Wiring shall be provided between motors, controllers, timers, pushbuttons, and switches installed on the equipment with materials specified in 3.3.4 as applicable. Flexible conduit shall be furnished where movement of a motor is required for belt adjustment. Installation of components shall conform to the requirements of NFPA Standard No. 70. All wiring shall terminate in connection boxes with provision for connections beyond the machine. Each conductor shall be identified by different colors. Wiring and connection diagrams shall be furnished with each tumbler.

3.4.1.1.3 Motor control circuit. All motors of 1 horsepower (746 W) or greater shall have a controller of the magnetic across-the-line type having a separate pole for each ungrounded conductor and conform to NEMA Standard ICS, as applicable. The tumbler shall have a timer, door interlock switch, indicating lights, buzzer and other controllers needed to perform the intended function. When the tumbler door is open the interlock switch shall automatically stop the tumbling action.

3.4.1.2 Safety guards. All belts, chains, shafts, pulleys, gears and other moving parts shall be fully inclosed or guarded. Guards shall be cast iron, sheet metal, or wire mesh. All parts of the guards shall be rigid and suitably secured and shall be readily removable without disassembling pipes or fittings on the machine. Provisions to permit easy access shall be made in guards fitted over parts requiring frequent adjustment with accesses fitted with covers.

3.4.1.3 Gearing. All gears shall have machined teeth. Spiral, hypoid, bevel, and worm gears, if used, shall operate in lubricant. When run in an oil bath, the pinion shall be of ferrous metal and the gear shall be of ferrous metal or bronze. Gears not run in an oil bath shall be of a phenolic formaldehyde resin or other noise-reducing material. When gears are of a composition material, either the motor pinion or gear with which it meshes shall be at least 1/4 inch (6 mm) wider than the other. When thrust bearings are used on the armature shaft the gear and pinion may be of the same width.

3.4.1.4 Belts and pulleys. Drives for equipment shall be through V-belts or chain. V-belts shall be of the grades commercially known as multiple-drive or variable speed types, except that where a single belt is furnished with a fractional-horsepower motor, a V-belt commercially known as a fractional-horsepower belt may be supplied in lieu of the multiple-drive or variable speed types. Pulleys for V-belts shall be suitably grooved. Chain drives shall be of the silent or roller-chain type. Means shall be provided for readily adjusting belts and chain tension.



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3.4.1.5 Piping system. All coils, pipes, fittings, and valves within the dry cleaning, recovery tumbler shall be installed in accordance with manufacturer's current commercial practice. Installed piping, except to chambers containing solvent or water open to the atmosphere, shall withstand without leaking, a 150-pound per square inch (1034 kPa) hydrostatic pressure when tested as specified in 4.3.2.

3.4.1.6 Seals. Means shall be provided to prevent the lubricant from entering parts of machines where clothes are handled. Means shall be provided to prevent cleaning solvent from entering the bearings.

3.4.1.7 Tumblers. Tumblers shall be a steam-heated, open-end, nonreversing type, and arranged to recirculate the hot air. The tumblers shall be equipped with condensing coils for reclaiming solvent; a timer, either a pressure or a temperature regulator; an automatic water cutoff, visual and audible signal to indicate that a drying cycle has been completed; steam pressure gage; thermometer; and safety latch (door interlock). Tumblers shall be designed to deliver heated air through the material being tumbled and evaporate the dry cleaning solvent and condense it back into liquid form. A water separator shall be provided. No leakage of water, solvent, oil or grease from the tumbler when tested as specified in 4.3.3. The tumblers shall be capable of drying their rated dry weight capacity of GFP woolen garments or blankets containing dry cleaning solvent at 25 percent or more of their dry weight to a maximum moisture content of 5 percent of their dry weight when tested as specified in 4.3.4. All tumblers shall be equipped with a deodorizing system containing a fan and exhaust damper arranged so that the fan operates continuously during loading and unloading of the tumbler and draws solvent vapors away from the operator and into the tumbler.

3.4.1.7.1 Frame. The tumbler frame shall be constructed of ferrous metal, formed to provide a rigid support for the housing, cylinder, heating and condensing coils, fan, and driving mechanism.

3.4.1.7.2 Housing. The tumbler housing shall be constructed of corrosion-resisting steel or other standard commercial ferrous sheet metal, coated to resist corrosion, of a minimum thickness of 0.0565 inch (1.44 mm). The housing shall be attached to the frame and shaped to insure effective air movement through the cylinder load. When separate housings are used for coils and are nonsupporting, the thickness of the metal for this housing shall be a minimum of 0.0272 inch (.69 mm). The tumbler housing shall encase the cylinder and shall be closed except for the exhaust outlet and the door opening. A tight-fitting door for convenient access to the interior of the cylinder for loading and unloading shall be provided in the housing. When the door is equipped with a window, the window shall be of heat-treated glass capable of resisting shock. The shell door shall be interlocked so as to disconnect power to the motor and cause the cylinder to stop when opened, see 4.3.3. All interior surfaces of the housing subject to contact with solvent vapors, when not of corrosion-resistant metal or commercial galvanized metal, shall be coated with epoxy baked enamel.



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3.4.1.7.3 Cylinder. The tumbler cylinder shall be open-end construction without door, and shall consist of a circular solid back sheet of corrosion-resisting steel or commercial galvanized ferrous metal not less than 0.0711 inch (1.81 mm) thick. Perforations shall be of sufficient number and size to allow passage of air through the cylinder without weakening side sheets to the extent that bulging or sagging occurs under capacity load. The gap between the open end of the cylinder and housing shall be covered to prevent loss or damage to the materials being processed. The cylinder shall have not less than three tumbling ribs, not less than 4 inches (102 mm) high with each rib fitted with a through tie rod to increase cylinder rigidity. The cylinder ribs shall be provided with replaceable pads made of canvas-covered felt or its equivalent.

3.4.2 Lubrication. Forced lubrication fittings shall be provided and accessibly located for lubricating bearings. Where lubricant reservoirs are required for chain or gear drives, they shall be provided with a threaded cap at the top and a drain plug at the bottom.

3.4.3 Marking for identification. Marking for identification shall be in accordance with the manufacturer's commercial practice.

3.4.4 Finish and color. The tumblers shall be finished in accordance with the manufacturers' standard practice. Emergency stop control push buttons shall be red in color.

3.4.5 Noise limits. The tumblers shall meet the noise limits specified in MIL-STD-1474 and where practicable, shall not exceed 85 decibels A-weighted at the normal operating position (see 4.3.5).

3.4.6 Electromagnetic compatability. When specified (see 6.2), equipment procured shall be designed and equipped to conform to class IIB of MIL-STD-461 (see 4.3.6).

3.4.7 Moisture and fungus resistant treatment. When specified (see 6.2), electrical components shall be moisture and fungus resistant treated in accordance with MIL-T-152.

3.4.8 Cylinder transmitter and trunnion. The circular back sheet shall be fasteneed to a commercial ferrous-metal transmitter having a trunnion rotating in ball or roller bearings.

3.4.9 Air exhaust. Openings shall be provided for exhausting the air passage through the tumbler and out the exhaust duct when the fan is operating. The openings shall be provided with duct connections.



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3.4.10 Lint trap. The tumbler shall have an accessible bag or screen for removing trapped lint.

3.4.11 Drive. The V-belt or roller chair drive shall consist of one or more motors to operate the cylinder and fan.

3.5 Contract data requirements. The contract data requirements shall be as specified (see 6.2 and 6.3).

3.6 Government-furnished property. The woolen garments or blankets shall be GFP (see 6.2).

3.7 Workmanship. The units, including inside surfaces of the cylinder, shall be free from defects such as fractures, splits, punctures, tears, dents, creases, deteriorations, or malformation. There shall be no sharp edges, slivers, burrs, or projections.

3.7.1 Application of finish. The finish applied to the end item shall be continuous, smooth, adherent, have no imbedded foreign matter and contain no sags, runs, drips, creeps, laps, bubbles or streaks. No rust or tool marks shall show through the coating.

3.7.2 Welding. The surface of parts to be welded shall be free from oxide, scale, paint, grease, and other foreign matter. Welds shall be sound, smooth, and free from porosity, cracks, incomplete fusion, and deformation of material. All scale and flux deposits shall be removed from the finished welds.

3.7.3 Soldering. Soldering shall be complete and adherent with all flux and residue removed and shall contain no pin holes.

3.7.4 Threaded fasteners. Threaded fasteners shall not be broken, stripped, fractured, or loose.

3.7.5 Rivets. Rivets shall completely fill the holes, with full size heads neatly finished and in full contact with the surface of the riveted members and shall not be loose.

3.7.6 Electrical wiring. Wiring shall not be cut, abraded, or have insulation stripped, and shall be joined at terminals. Wiring shall have adequate slack to provide strain relief.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance



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of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Except as otherwise specified herein sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified, or qualified in this specification.

4.2.2 End item inspection. The lot shall be all tumblers of one type, size and style as applicable, offered for inspection at one time. The sample unit shall be one recovery tumbler.

4.2.2.1 Visual examination. The completely assembled item shall be examined for defects in table I. The inspection level shall be II and the acceptable quality level (AQL) shall be 2.5 for major and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

| Examine | Defect | Classification Major | Minor |
|---|---|---|---|
| Finish | Not finished where required | X | |
| | Finish and color not as specified | X | |
| | Not continuous, smooth, adherent, foreign matter imbedded and sags, runs, drips, creeps, laps, bubbles or streaks | | X |
| | Rust or tool marks | | X |
| Construction | Any part missing or not specified type | X | |
| | Split, punctured, torn, dented, creased, deteriorated or malformed | X | |
| | Sharp burr, sliver or sharp edge | X | |
| | Not lubricated where required | | X |
| | Excessive lubrication | | X |



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TABLE I. Classification of defects (cont'd)

| Examine | Defect | Classification Major | Minor |
|---|---|---|---|
| Marking | Not manufacturers commercial practice | | X |
| Welding and brazing (where applicable) | Missing, incomplete, burn holes, fractured, or not fused | X | |
| | Slag inclusion, undercut, not smooth and uniform, or deposit of scale and flux not removed | X | |
| Soldering (where applicable) | Missing, incomplete or not adherent | X | |
| | Not clean (flux or residue not removed) or contains pinholes | | X |
| Bolts, nuts, screws studs and other threaded fasteners | Missing, broken, stripped or fractured | X | |
| | Threaded fasteners loose | | X |
| Rivets (where applicable) | Missing, broken, not peened or not peened sufficiently | X | |
| | Rivets loose | | X |
| Electrical apparatus wiring | Wiring cut, abraded, insulation stripped, not properly joined or loose at terminals | X | |
| | Not inclosed in specified conduit where required | X | |
| | Adequate slack not provided for wiring to relieve strain | | X |
| Marking for identification | Missing, incomplete, not legible | | X |

4.2.2.2 Dimensional examination. Inspection shall be made of the end item for compliance with dimensions specified. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units. Any nonconformance shall be classified a defect.

4.2.2.3 Testing of the end item. A random sample from the first production lot shall be tested as specified in 4.3.3, 4.3.4, 4.3.6, and when applicable 4.3.5. Any nonconformance shall be cause for rejection of the random sample.



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4.2.2.4 Code and standards compliance. Proof of compliance with the requirements of 3.2.1, 3.2.2, and 3.2.3 shall be made available to the Government representative. When moisture and fungus resistant treatment is specified, the contractor shall furnish a certificate of compliance that electrical components comply with 3.4.1.13.

4.2.3 Inspection of preparation for delivery. Preservation-packaging, packing and marking shall be inspected in accordance with the quality assurance provisions in MIL-L-3153.

4.3 Tests.

4.3.1 Operational testing. Each tumbler shall be connected to power and operated without load and solvent. Nonconformance to 3.4.1 constitutes failure of this test.

4.3.2 Hydrostatic pressure tests. Assemblies of pipes, fittings, coils, cylinders, and chambers using steam, water or solvent, except those open to the atmosphere, shall be tested with 150 pounds per square inch (1034 kPa) hydrostatic pressure. The test pressure shall be maintained a minimum of 10 minutes for each assembly to determine that no leaks are present. Nonconformance to 3.4.1.5 constitutes failure of this test.

4.3.3 Functional testing. Each tumbler shall be tested and operated under full load requirements (GFP garments or blankets) with solvent as specified in 3.4.1.7. The following shall checked and nonconformance shall be cause for rejection:

(a) Lubricant seals for compliance with the requirements of 3.4.1.6.
(b) Leakage of parts for compliance with the requirements of 3.4.1.7.
(c) Tumbler shell door interlocking device for compliance with the requirements of 3.4.1.7.2.
(d) Means of lubrication for compliance with the requirements of 3.4.2.

4.3.4 Tumbler drying capacity test. The tumbler drying capacity test shall be performed by placing in each tumbler the rate capacity of woolen garments or blankets containing dry cleaning solvent at 25 percent, or more, of their "dry" weight. With the circulating air in the tumbler heated to not more than 150°F (65.5°C), the woolens shall be tumbled for not more than 15 minutes and removed from the tumbler and weighed. Nonconformance to 3.4.1.7 shall constitute failure of this test.

4.3.5 Noise test. The tumbler shall be tested for conformance with the noise limits of 3.4.5 in accordance with MIL-STD-1474.



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4.3.6 Electromagnetic compatability test. When electromagnetic compatability is required, the initial unit of production shall be tested by the contractor in accordance with methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the contractor or an independent testing agency. The contractor shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the EMI/EMC Test Report Plan, the electromagnetic compatability test report and the requirements meet MIL-STD-461 (see 3.4.6).

5. PACKAGING

5.1 Preservation-packaging, packing and marking. Preservation-packaging, packing and marking for level A, B or C shall be in accordance with MIL-L-3153 (see 6.2).

6. NOTES

6.1 Intended use. The tumblers covered by this specification are intended for use with a separate washer-extractor, filter, a combination sludge cooker and still to form a complete closed or transfer system for using synthetic solvent, nonflammable at ordinary temperatures.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) Electrical characteristics (voltage, a.c. or d.c.; frequency, and number of phases required (see 3.4.1.1.1).
(d) When electromagnetic compatibility is required (see 3.4.6).
(e) When moisture and fungus-resistant treatment is required (see 3.4.7).
(f) Contract data requirements (see 3.5 and 6.3).
(g) Government-furnished property (see 3.6 and 6.5).
(h) Selection of applicable levels of preservation-packaging and packing (see 5.1).

6.3 Contract data requirements. Any data items to be delivered under any contract for items covered by this specification shall be specifically called for in the contract in accordance with the applicable regulations of the procuring agency (see 3.5 and 6.2).

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).



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6.5 Government furnished property. The contracting officer should arrange to furnish the property specified in 3.6.

6.6 Supersession data. The 35 and 50 pound capacity tumblers of this specification replace the 40 and 50 pound capacity tumblers in MIL-D-43164B, respectively. The 70 pound capacity tumbler of MIL-D-43164B is no longer required.

Custodians:

Army - GL
Air Force - 99

Review activities:

Army - MD
Air Force - 45
DLA - GS

Preparing activity:

Army - GL

Project No. 3510-0269



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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-D-43970 -- Dry Cleaning, Recovery Tumbler, For Synthetic Solvent

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

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